

Analysis of the LQR Controller

1 The LQR Controller

1.1 How LQR Works

The LQR controller computes an optimal state-feedback gain K by minimizing the cost function:

$$J = \int_0^\infty (x(t)^\top Q x(t) + u(t)^\top R u(t)) dt,$$

where:

- Q is a weighting matrix penalizing deviations in the state variables.
- R is a scalar (or matrix) penalizing the control effort.

The feedback control law is:

$$u = -Kx.$$

This means that the control input u is proportional to the state x (with a negative sign to provide corrective action).

1.2 Weighting Matrices Q and R

- $Q = \text{diag}([100, 1, 1000, 1])$:
 - High weights (100 and 1000) on the cart position and pendulum angle indicate that errors in these states are highly undesirable.
 - Lower weights (1) on the velocities suggest that while velocity errors are important, they are less critical than position or angle errors.
 - **Effect of Changing Q :** Increasing an element of Q forces the controller to be more aggressive in correcting that particular state, which can reduce rise and settling times but may lead to higher control efforts.
- $R = 0.01$:
 - A smaller R implies that the control effort is less penalized, allowing for stronger control actions.
 - **Effect of Changing R :** Increasing R penalizes the control input more, leading to a less aggressive controller. This can reduce energy consumption but may result in a slower response.

1.3 Performance Impact and Trade-offs

The controller's performance is influenced by the relative magnitudes of Q and R :

- **Aggressiveness vs. Energy Efficiency:** A higher Q relative to R results in a more aggressive control action, reducing rise and settling times but possibly causing actuator saturation and higher energy consumption.
- **Stability and Robustness:** The closed-loop system is given by:

$$A_{cl} = A - BK.$$

The eigenvalues of A_{cl} indicate the system's stability. Tuning Q and R changes these eigenvalues, thereby affecting transient behavior such as damping, overshoot, and settling time.

- **Trade-offs:**
 - *Too aggressive (high Q and/or low R):* May lead to fast response but can cause actuator saturation and potential instability.
 - *Too conservative (low Q and/or high R):* Can result in a sluggish response, longer settling times, and increased risk of the pendulum falling before adequate correction.

2 Simulation and Analysis

The provided MATLAB code performs several tasks:

1. **Gain Computation:** It computes the optimal gain K using the LQR algorithm.
2. **Closed-Loop Simulation:** Using the state-space model $A_{cl} = A - BK$, the response from an initial condition $x_0 = [0; 0; 0.1; 0]$ is simulated.
3. **Transient Response Analysis:** Metrics such as rise time and settling time are determined via the `stepinfo` function, showing how quickly and effectively the controller stabilizes the pendulum.
4. **Frequency-Domain Analysis:** The root locus, Nyquist, and Bode plots provide insights into stability margins, phase/gain margins, and overall robustness.

3 Summary

- The LQR controller is designed to balance the pendulum by minimizing the cost function that balances state deviations (via Q) and control efforts (via R).

- Physical parameters ($m_w, m_p, I_w, I_p, l, r, g$) and intermediate coefficients (a, b, c, d, Δ) encapsulate the system dynamics.
- Tuning Q and R directly impacts the closed-loop behavior:
 - A higher Q (or lower R) results in a more aggressive control response.
 - A lower Q (or higher R) results in a more conservative, energy-efficient response.
- The eigenvalues of the closed-loop system ($A - BK$) reveal the stability and transient characteristics, guiding further tuning of the controller.